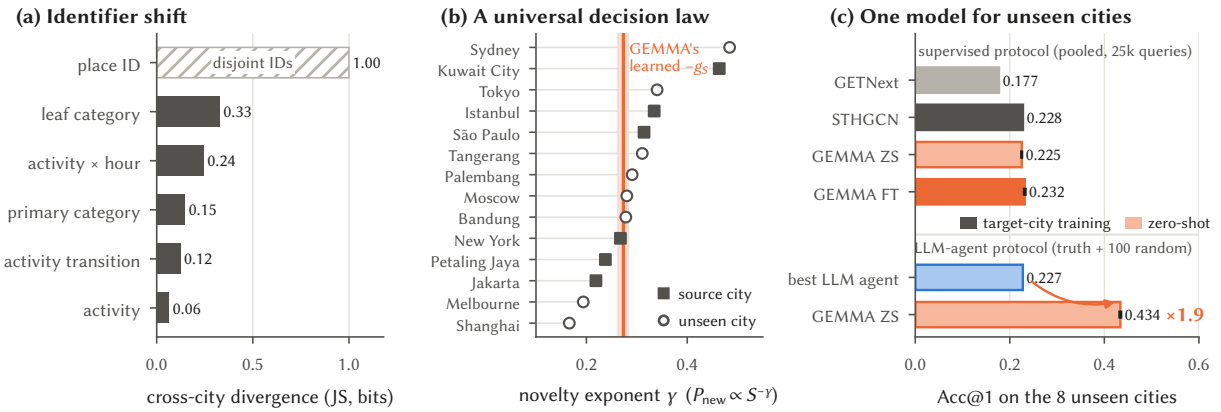
(a) Identifier shift
place ID
disjoint IDs
1.00
leaf category
0.33
activity × hour
0.24
primary category
0.15
activity transition
0.12
activity
0.06
0.0
0.5
1.0
cross-city divergence (JS, bits)
(b) A universal decision law
Sydney
Kuwait City
Tokyo
Istanbul
São Paulo
Tangerang
Palembang
Moscow
Bandung
New York
Petaling Jaya
Jakarta
Melbourne
Shanghai
GEMMA's learned $-g_S$
source city
unseen city
0.2
0.4
novelty exponent γ ($P_{\text{new}} \propto S^{-\gamma}$)
(c) One model for unseen cities
supervised protocol (pooled, 25k queries)
GETNext
0.177
STHGCN
0.228
GEMMA ZS
0.225
GEMMA FT
0.232
target-city training
zero-shot
LLM-agent protocol (truth + 100 random)
best LLM agent
0.227
GEMMA ZS
0.434
×1.9
0.0
0.2
0.4
0.6
Acc@1 on the 8 unseen cities